Abdul Khurram

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Education & Awards

University of British Columbia — Vancouver, BC, Canada
*Bachelor of Science in Computer Science (**Expected Graduation: May 2027**)*

Euclid Mathematics Contest, University of Waterloo — 2021
Top 5% Internationally, Certificate of Distinction

Work Experience

RBC Royal Bank | ***Python, Apache Spark, Airflow****, SQL, Linux* — Jan. 2026 – Present
Software Developer Co-op — *Vancouver, BC, Canada*

- Engineered **Python**/Airflow **ETL pipelines** powering **ML** workloads across 50M+ daily transactions on petabyte-scale Delta Lake, delivering feature-ready datasets for **model training** and real-time inference.
- Optimized **Spark algorithms** via Spark-native API migration and AQE/cluster-memory tuning, cutting latency 40% and compute costs through **distributed computing** resource profiling.
- Hardened batch processing framework with observability, fault tolerance, and automated failover; shipping cost/performance dashboards driving real-time infrastructure decisions across the platform.

Contello.ai | ***Python****, TypeScript/Node.js, AWS, OpenAI* — Jan. – Dec. 2025
Software Engineer Intern — *Toronto, ON, Canada*

- Integrated OpenAI Responses API into production TypeScript/Node.js on AWS; built content-generation API services serving thousands/day with **model inference** observability via CloudWatch.
- Enabled **ML**-assisted content creation by ingesting brand docs via an **NLP** pipeline, caching **embeddings**, and auto-injecting context while enforcing responsible AI guardrails.
- Raised service reliability to 99.9% via **model** API migration, cost/latency tracking, retry/backoff, and automated tests in CI/CD; deployed via Docker containers on AWS.

Research Experience

UBC Emerging Media Lab | *Unreal Engine, **C++**, **Python*** — Sept. – Dec. 2024
Software Developer & Team Lead — *Vancouver, BC, Canada*

- Led a **human-AI collaboration research** project: built an AI-driven VR simulation in Unreal (**C++**) pairing nurse practitioners with AI patients for clinical training.
- Designed a multithreaded supervisory architecture (**C++/Python**) enabling real-time **ML model** monitoring and protocol enforcement across autonomous patient actors.
- **Optimized** inference pipeline, achieving 40% latency reduction and 90% cost savings through profiling and **algorithm** tuning; documented design decisions and system capabilities.

Projects

CircleFund | ***Python, PyTorch****, TypeScript/Next.js, Prisma/PostgreSQL* — **Personal Project**

- Building a full-stack lending platform (Next.js, Prisma/PostgreSQL) with account onboarding, invite-code circle management, and contribution dashboards for community-group financial workflows.
- Designing **PyTorch**-based borrower risk scoring and contribution reliability **models** with planned liquidity stress forecasting, integrating **ML** predictions into the platform's decision layer.

Backer | *TypeScript/Next.js, AWS DynamoDB, NextAuth* — **Hack The Coast 2026**

- Architected a startup discovery platform (TypeScript/Next.js, DynamoDB) serving personalized investor feeds at scale via API services.
- Built **ML** training pipeline: supervised **model** with sequence and clustering features, ingesting signals from DynamoDB event streams to train and deploy an **XGBoost** ranker to production.

Technical Skills

Languages: **Python**, Java, **SQL**, TypeScript/JavaScript, **C/C++**, Bash, HTML/CSS
ML & Platforms: **PyTorch**, **Scikit-learn**, OpenAI API, LangChain, **NLP**/embeddings; React/Next.js, **Airflow**, **Hadoop/Spark**; AWS (Lambda, **DynamoDB**, S3, SQS, CloudWatch); **PostgreSQL**, Docker, Git, CI/CD